Question 1 (15 marks)

Show your workings clearly and round off your answers to 1 decimal place if necessary.

A. Tom is going overseas to work for a period. To budget for accommodation, he gathered some data on the average rental price of a small apartment in the city (Table 1a).

Table 1a: Average rental price of small apartment in last 6 months

Month	Average Rental Price
Jan	$550
Feb	$480
Mar	$490
Apr	$510
May	$470
Jun	$460

a) Compute the forecast of the average rental price in July based on 5-period moving average. (2 marks)

(480+490+510+470+460)/5 = 482

b) Comment on the impact to the forecast if a longer moving average period (e.g. 12 months) is used. (2 marks)

As N increases, the forecast becomes less responsive to changes in actual data and the graph becomes smoother.

a) Determine the number of periods the forecast will lag if Tom continues to forecast using 5-period moving average and there is a trend in the rental price going forward. (2 marks)

MA = (N+1)/2
= (5+1)/2
= 3

b) What forecasting method would you suggest Tom to use if he wants his forecast to reflect more recent rental price data? (2 marks)

Weighted Moving Average (WMA)

B. As Tom would be out of Singapore for more than a year, he decided to rent out his HDB flat. He pulled out a graph on HDB rental price index (Figure 1a).

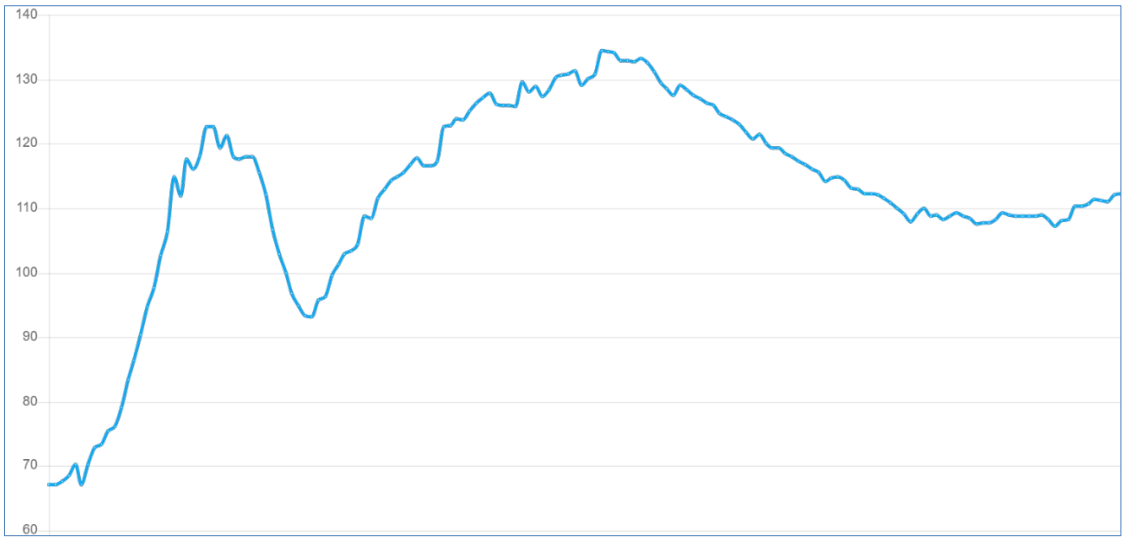

Figure 1a: HDB rental price index from Jan 2006 to Jul 2019

a) State **2** components of the time series that can be observed in Figure 1a. (2 marks) Random & Cyclicity

b) Tom used the last few data points from the graph and derived the selling price of his flat over the last 3 periods. See Table 1b for his calculations using Holt's method. Assume Alpha = 0.3 and Beta = 0.5.

Table 1b: Holt's method used to forecast selling price

Period	Selling Price ($'000)	Level (a)	Trend (b)	Forecast (c)
…	…	…	…	…
8	312	309.2	8.6	308.0
9	323	319.4	9.4	317.8
10	327	328.26	9.13	328.8

i) Calculate the price forecast for **period 10**. (2 marks)

H10 = 319.4 + 9.4 = 328.8

ii) Calculate the price forecast for **period 12**. (3 marks)

L10 = (0.3)(327)+(1-0.3)(319.4+9.4) = 328.26
T10 = (0.5)(328.26-319.4)+(1-0.5)(9.4) = 9.13

H12 = L10 + 2(T10)
= 328.26 + 2(9.13)
= 346.52